# IGCSE 0607 Practice Questions — E1.4

## Fractions, decimals and percentages

**Total Questions:** 12 — **Total Marks:** 68 Marks

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## Table of Contents & Quick Navigation


- **E1.4** Fractions, decimals and percentages
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*


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**1** **(a)** Write $0.\dot{7}$ as a fraction.

..... **[1]**

**(b)** Write $0.\dot{3}\dot{6}$ as a fraction in its simplest form.

..... **[2]**

**2** **(a)** Write 45% as a fraction in its simplest form.

..... **[1]**

**(b)** Write $\frac{9}{25}$ as a decimal.

..... **[1]**

**(c)** Write 2.04 as a mixed number in its simplest form.

..... **[2]**

**3** **(a)** Write 0.45 as a fraction in its simplest form.

..... **[1]**

**(b)** Write $\frac{3}{8}$ as a decimal.

..... **[1]**

**(c)** Write these in order of size, starting with the smallest.

$$\frac{5}{12} \qquad 0.4 \qquad 42\% \qquad \frac{3}{7}$$

..... **[2]**

**4** **(a)** Write 0.075 as a fraction in its simplest form.

..... **[2]**

**(b)** Write $3\frac{2}{5}$ as a decimal.

..... **[1]**

**(c)** Write the following in order of size, starting with the smallest.

$$62\% \qquad \frac{5}{8} \qquad 0.6 \qquad \frac{7}{11}$$

..... **[2]**

**5** The length of a rectangle is increased by $33.\dot{3}\%$.
The width of the rectangle is decreased by $x\%$.
The overall area of the rectangle increases by $16.\dot{6}\%$.
Calculate the value of $x$.

$x =$ ..... **[5]**

**6** A bag contains red counters, blue counters and green counters.
The number of red counters is $x$.
The number of blue counters is $2x + 5$.
The number of green counters is $3x - 1$.

**(a)** Exactly 35% of the counters in the bag are blue.
Work out the value of $x$.

$x =$ ..... **[4]**

**(b)** Work out the percentage of the counters in the bag that are green.
Write your answer as a mixed number.

..... % **[3]**

**7** **(a)** Write the recurring decimal $0.2\dot{4}\dot{5}$ as a fraction in its simplest form.
You must show all your working.

..... **[3]**

**(b)** The fraction $\frac{N}{42}$, where $N$ is a positive integer, is in its simplest form.
Explain why the decimal equivalent of $\frac{N}{42}$ is a recurring decimal.

.....

..... **[2]**

**(c)** The recurring decimal $0.\dot{x}\dot{y}$, where $x$ and $y$ are distinct digits, can be written as a fraction $\frac{k}{11}$, where $k$ is an integer.
Find the value of $x + y$.

$x + y =$ ..... **[2]**

**8** **(a)** By converting the recurring decimal to a fraction, show that

$$0.2\dot{7} = \frac{5}{18}$$

**[2]**

**(b)** In a school, $\frac{2}{5}$ of the students are boys and the rest are girls.
45% of the boys play a sport.
30% of the girls play a sport.

**(i)** Work out the percentage of all students in the school who play a sport.

..... % **[3]**

**(ii)** There are 144 students who play a sport.
Work out the total number of students in the school.

..... **[2]**

**9** In a school, $\frac{3}{8}$ of the students are in the Junior section, 45% are in the Middle section, and the rest are in the Senior section.
There are 140 students in the Senior section.

**(a)** Work out the total number of students in the school.

..... **[3]**

**(b)** The number of boys in the Junior section is 150% of the number of girls in the Junior section.
Work out the number of girls in the Junior section.

..... **[3]**

**10** **(a)** Write the following in order of size, starting with the smallest.

$$\frac{5}{7} \qquad 71\% \qquad 0.714 \qquad \frac{3}{4}$$

Smallest ..... **[2]**

**(b)** Find the fraction which is exactly halfway between $\frac{2}{5}$ and $\frac{5}{8}$.
Give your answer in its simplest form.

..... **[2]**

**(c)** A fraction is equivalent to $\frac{4}{9}$.
The difference between the denominator and the numerator is 35.
Find the fraction.

..... **[2]**

**11** A company has three departments: Sales, Production, and Administration.

- $\frac{2}{5}$ of the employees are in Sales.
- 28% of the employees are in Production.
- The remaining 24 employees are in Administration.

**(a)** Work out the total number of employees in the company.

..... **[3]**

**(b)** 60% of the Sales employees are women.
$\frac{4}{7}$ of the Production employees are men.
Half of the Administration employees are women.
Work out the percentage of the whole company who are women.

..... % **[4]**

**12** **(a)** A television costs \$450. In a sale, the price is reduced by 18%.
Calculate the sale price of the television.

\$ ..... **[2]**

**(b)** In the same sale, a laptop is sold for \$629 after a 15% reduction.
Calculate the original price of the laptop.

\$ ..... **[3]**

**(c)** The price of a tablet is increased by $\frac{1}{5}$.
Later, this new price is decreased by 20%.
Calculate the overall percentage decrease in the price of the tablet.

..... % **[2]**